Water Management As a New Saving Opportunity for Bus Managers: Evidence from the LIFEH2OBUS Project

Matthew Robinson and Maria Vittoria Corazza(✉)

Sapienza University of Rome, Rome 00184, Italy
{matthew.robinson,Mariavittoria.corazza}@uniroma1.it

Abstract. LIFEH2OBUS, a project within the European Commission's LIFE program, aims to evaluate water usage in the bus sector during vehicle cleaning. This research is pioneering because water conservation in this field remains largely unexplored, lacking specific standards to prevent unnecessary water consumption. A single bus cleaning session requires around 300 litres of water, occurring four times a week per bus, totalling 43 million cubic meters annually for the European bus fleet. Furthermore, bus washing is energy-intensive, and most garages lack wastewater treatment facilities. LIFEH2OBUS and this paper assess the implementation of three innovative water-saving technologies in bus garages across Europe: water reclamation, water reclamation and harvesting, and waxing. A cost-benefit analysis reveals an impressive 84% reduction in water consumption after one year, saving 37 million litres for the test fleet of 680 buses. Scaling up to 50% of the European transit fleet over five years could save 18 billion litres/year, equivalent to 42% of the transport sector's water use, reduce energy consumption by 1,159 GWh, and cut 504 ktCO_2eq greenhouse gas emissions, saving 151 million Euros. This paper introduces this technological process and demonstrates its substantial water and energy-saving potential in the transport sector, advancing scientific knowledge in this field.

Keywords: Water · Bus · LIFEH2OBus · Water Saving

1 Introduction

The European bus fleet industry has been driven by a strong commitment to environmental sustainability, primarily focusing on reducing pollution and resource consumption. Over the past few decades, considerable Research and Development (R&D) efforts have been dedicated to shaping the vision of the "European bus of the future." Financial support from the European Union (EU) has been instrumental in facilitating successful research projects since the early 2000s. Notably, energy management has emerged as a key area of innovation, encompassing electrification, cleaner engine technologies, and alternative fuels [1, 2]. Several factors have propelled these R&D endeavours, including uncertainties in energy supply [3], increasingly stringent European environmental regulations, and the growing consensus on green economy strategies [4]. A particular

© The Author(s) 2026
C. McNally et al. (Eds.): TRAconference 2024, LNMOB, pp. 586–592, 2026.
https://doi.org/10.1007/978-3-032-04774-8_85

focal point is the development of vehicle components and systems aimed at reducing emissions and energy consumption. Numerous studies have underscored the advantages of such innovations, including heightened productivity, improved reliability, optimized resource utilization, cost savings, and environmentally conscious performance [5, 6]. Consequently, transport operators are often driven to innovate to achieve cost savings and comply with stricter environmental standards. Strikingly, water management and consumption have remained a relatively neglected area, not only among transport operators but also among policymakers and researchers. The scarcity of data, recommendations, guidelines, or organized information at the EU level regarding water usage in the public transport sector indicates this oversight. Moreover, the existing transportation literature offers limited insights into this domain, with a conspicuous absence of comprehensive contributions beyond informal sources. To address this gap, the EU-funded project LIFEH2OBUS, focused on promoting best practices for water management and savings in the bus industry, seeks to evaluate water-saving technologies at three European bus garages. This paper thoroughly examines the processes and outcomes achieved at one of these garages, highlighting the substantial potential for water-related savings. These findings could catalyze additional case studies and further advancements in scientific understanding.

2 Understanding the Problem

Globally, water scarcity has become a pressing issue, with reported stress situations worldwide. According to the European Environmental Agency, about 100 million Europeans face water stress, exacerbated by declining renewable water availability, partly due to climate change. Agriculture accounts for 40% of annual water use, followed by energy production at 28%, with households requiring 12% [7]. Remarkably, washing a single bus consumes approximately 300 litres of freshwater three times a week [8], resulting in an annual water requirement of approximately 46,800 litres per vehicle. When scaled up to encompass the entire European bus fleet, consisting of 684,285 vehicles [9], this equates to a staggering total of around 32 million cubic meters of freshwater needed for washing operations. While commercial literature is abundant online regarding various bus washing and wastewater systems, scientific literature analyses water consumption within the broader context of car washing operations, with few studies explicitly focusing on the bus sector [10, 11]. In contrast, there is more substantial interest in water treatment practices and technologies, although not always directly related to transit operations. The environmental [12, 13] and economic [14] implications of wastewater treatments are discussed without focusing on the transit sector. These observations underscore the pressing need for specialized knowledge regarding sustainable water management practices for bus operators, especially in light of policies aimed at achieving more sustainable transportation, such as the European Green Deal's goal of reducing greenhouse gas emissions in the transportation sector by 90% by 2050. The COVID-19 pandemic has also brought cleanliness in travel choices into focus [15]. However, much of the analysis has centered on personal protection equipment and physical distancing [16], with bus washing remaining largely unexplored. This aspect is particularly pertinent as increased fresh air intakes on buses might offset previous energy-saving efforts [1, 16], emphasizing the

continued significance of water for sanitization. Moreover, transit cleaning standards often depend on local tender agreements, leading to inconsistent practices and inadequate wastewater treatment, often directly discharging highly polluting wastewater into the sewage systems. The energy required for these processes is also substantial, with data indicating that 6.60 kWh are needed to pump 1 m^3 of water [8]. This results in an annual energy requirement of around 211.36 GWh for washing the European bus fleet. Regarding CO_2eq emissions, approximately 0.005 t CO_2eq [17] is generated for each cubic meter of water pumped, resulting in roughly 160,122.69 t CO_2eq emitted annually by the bus sector solely for washing operations. Budget constraints and financial considerations inherent in bus operations make energy savings crucial. Water, as demonstrated, can no longer be considered a minor cost item, especially as water costs in Europe have escalated over the past decade, and the price of sewerage and wastewater services has followed suit [18].

3 Understanding the Problem

All of the above also highlights a significant gap between policy development and current knowledge and practices, which the LIFEH2OBUS project aims to fill by providing bus operators with practical insights into the benefits of recycling wastewater during bus washing and/or adopting waterless methods. To assess potential savings in energy, operations, and environmental impact, a do-something scenario lasting 30 months was developed, simulating the introduction of three distinct technologies across three European garages (in the Mediterranean, Western, and Central Europe regions, to cope with specific weather conditions, i.e.). These are:

- Basic reclamation involves a wastewater system comprising a 1,000-liter oil separator, a coalescent filter, and a 5,500-liter post-sedimentation tank. A water pump facilitates the removal of oils and suspended solids. Further filtration and disinfection occur in a recirculation section with a sand filter. The cleaned water is stored in a 2,000-liter tank, ready for subsequent washing operations.
- Reclamation and harvesting, combining wastewater treatment and recycling with a rainwater harvesting system.
- Waxing without the use of water, as for aircraft.

All were selected to address the research question of reducing water consumption during washing operations while maintaining cost-effectiveness and scalability.

The scenario fleet consists of 680 buses operated by Arriva (the LIFEH2OBUS coordinator) and corresponds to the fleets that are really in use at each garage. The do-something scenario was created without modifying the service operations, thus maintaining the washing schedule as four times a week for each vehicle (300 litres of water per operation). Cost calculations for the three technologies are based on market conditions and regional prices. Expenses encompass the construction of all the infrastructure, equipment, and electrical and control software units. These costs range from 250,000 to 290,000 euros. Basic reclamation equipment costs approximately 110,000 euros, with the water re-rotation system and scissor bus lift being the pricier components (53,000 euros). Site preparation and design, including hydraulics and groundwork, range from

50,000 to 70,000 euros. For the waterless waxing procedure, each operation costs 250 euros. The scenario envisions ten operations per year per bus, resulting in an annual cost of 187,000 euros for the 75 vehicles across the three garages. Lastly, training the maintenance team to operate the new equipment is anticipated to require an average expenditure of around 10,000 euros.

4 The Saving Potential

According to the conditions above reported, the performance comparison between the do-something scenario vs. the reference practice shows, at the three test sites, that it is possible to reduce water consumption, and most specifically, the weekly freshwater intake passes from 1200 litres to 480 litres (basic reclamation), 96 litres (reclamation and harvesting), and finally to 24 litres (when waxing). Contrarily, the majority of the water is supplied by processed wastewater and rainwater. The cumulative impact of having two technologies in a single garage generates a potential saving of almost 37 million litres of freshwater in the three test sites. This means an average of 84% due to the deployment of the three technologies (Table 1), with reclamation and harvesting delivering a 92% reduction of freshwater intake, basic reclamation only enabling 60 and waxing being the most water-saving approach (−98%).

Table 1. The combined effect of technologies on water-saving.

Technology	Scenario	Performance	Test areas		
			Mediterranean	Western Europe	Central Europe
Reclamation and harvesting	Reference	Consumed freshwater by garage fleet (l/y)	12480000	21840000	
	Do-something	Consumed freshwater by test fleet (l/y)	873600	1622400	
Basic reclamation	Reference	Consumed freshwater by garage fleet (l/y)			8112000
	Do-something	Consumed freshwater by test fleet (l/y)			2620800
Waxing	Do-something	Consumed freshwater by test fleet (l/y)	31200	31200	31200

(*continued*)

Table 1. (*continued*)

Technology	Scenario	Performance	Test areas		
			Mediterranean	Western Europe	Central Europe
Total water consumed (l/y)			904800	1653600	2652000
Water consumption reduction (l/y)			−11575200	−20186400	−5460000
Water consumption reduction (%)			93	92	67

Table 2. Monetary savings and pollution reduction (30-month do something scenario).

Performance	Test areas			
	Mediterranean	Western Europe	Central Europe	Total
Water consumption reduction (l/y)	−11575200	−20186400	−5460000	−37221600
Energy consumption reduction (kWh)	−743128	−1295967	−350532	−2389627
Pollution reduction - GHG (tCO_2eq)*	−338123	−582665	−159492	−1087280
Savings (Euro)	−321	−561	−152	−1034

*assumed rate: 0.02777 kWh/m^3 [18]

The monetary conversion of these results is computed on assumptions derived from real-world situations. Local expenses for water and energy may vary, but Arriva's analysis of its depots indicates that, on average, the cost of water usage is 0.13 Euros per cubic meter. When considering the EU's average energy price during the first half of 2021, which was 0.1283 Euro per kWh [19], each cubic meter of water pumped for bus washing incurs a cost of 8.24 Euros. Pumping one cubic meter of water demands 64.2 kWh. Consequently, the total cost of water for the entire washing process averages 8.37 Euros per cubic meter. Utilizing the EPA model [17], Table 2 calculates that the 84% reduction in water wastage at the three test locations translates to savings exceeding 300,000 euros and an emission reduction of over 1000 tCO_2eq.

5 Concluding Remarks

Figures in Tables 1 and 2 may seem modest at first glance, but their implications become evident when applied to Europe on a larger scale, even with a conservative upscaling scenario. The 43 million cubic meters of freshwater needed to wash the European 684,285 buses result in an annual energy consumption of 2.77 TeraWh and the emission of 1.2 $MtCO_2eq$ of greenhouse gases, assuming consistent maintenance standards for the entire European bus fleet. Scaling up the savings potential calculated for the three test areas

and projecting a 50% adoption rate in the European bus transport sector within five years (equivalent to 342,143 buses), 18 billion litres of water annually can be saved, a substantial −42% reduction in total water usage in the transportation sector. This also leads to a 1,151 GWh reduction in energy consumption and a significant −496 $ktCO_2eq$ decrease in greenhouse gas emissions, resulting in a total cost reduction of 150 million Euros. To put this into context, this cost reduction could finance the purchase of more than 230 modern 12-m electric buses, priced at around 650,000 euros each, or potentially double that number for standard EURO-IV models. These insights serve as a sustainability benchmark, and results prove to meet environmental requirements and resource-saving objectives; consequently, bus operators may find transitioning from conventional washing systems to innovative washing technologies a viable and less expensive alternative to fleet renewal.

References

1. Musso, A., Corazza, M.V.: The European bus system of the future: research and innovation. Transp. Res. Procedia. **5**, 13–29 (2015)
2. Musso, A., Corazza, M.V.: Visioning the bus system of the future: stakeholders' perspective. Transp. Res. Rec. **2533**, 109–117 (2015)
3. Johnstone, P., McLeish, C.: World wars and sociotechnical change in energy, food, and transport: a deep transitions perspective. Technol. Forecast. Soc. Chang. **174**, 121206 (2022)
4. Moretti, L., Di Mascio, P., D'Andrea, A.: Green public procurement criteria for road infrastructures: state of the art and proposal of a weighted sum multicriteria analysis to assess environmental impacts. ARPN J Eng Applie Sciences. **12**(17), 4887–4894 (2017)
5. Campos, M., Mensión, J., Estrada, M.: Charging operations in battery electric bus systems at the depot. Transp Res Proca. **58**, 103–110 (2021)
6. López, C., Ruíz-Benítez, R., Vargas-Machuca, C.: On the environmental and social sustainability of technological innovations in urban bus transport: the EU case. Sustainability. **11**(5), 1413 (2019)
7. European Environmental Agency, Water use in Europe. Quantity and quality face big challenges, https://www.eea.europa.eu/signals/signals-2018-content-list/articles/water-use-in-europe-2014, last accessed 2021/12/12.
8. Arriva. ANEYT-EP-8.1.2 Operational Planning and Control - Water Management, Internal document (2019), unpublished.
9. Automobile Manufacturers' Association. Vehicles in use, Europe 2022. https://www.acea.auto/publication/report-vehicles-in-use-europe-2022/, last accessed 2022/12/12.
10. Almeida, C.M.V.B., Borges, D., Bonilla, S.H., Giannetti, B.F.: Identifying improvements in water management of bus-washing stations in Brazil. Resour. Conserv. Recycl. **54**(11), 821–831 (2010)
11. Coombes, P.J., Argue, J.R., Kuczera, G.: Figtree place: a case study in water sensitive urban development (WSUD). Urban Water. **1**(4), 335–343 (2010)
12. Rosa, L.G., Sousa, J.T., Lima, V.L., Araujo, G.H., Silva, L.V., Leite, V.D.: Characterisation of wastewaters from vehicle washing companies and environmental impacts. Ambiente E Agua - An Interdisciplinary Journal of Applied Science. **6**, 179–199 (2011)
13. Dadebo, D., Ibrahim, M.G., Fujii, M., Nasr, M.: Transition towards sustainable carwash wastewater management: trends and enabling Technologies at Global Scale. Sustainability. **14**(9), 5652 (2022)
14. Davis, L.: 5 tips on buying the ideal bus washing system. Metro. **98**(6), A24–A29 (2002)

15. Shortall, R., Mouter, N., Van Wee, B.: COVID-19 passenger transport measures and their impacts. Transp. Rev. **42**(4), 441–466 (2022)
16. Corazza, M.V., Emberger, G., Oszter, V., Pejdo, A., Shibayama, T.: COVID-19 containment measures and policies for sustainable transport: present changes and future directions. Ingegneria Ferroviaria. **76**(9), 669–697 (2021)
17. Environmental Protection Agency. Greenhouse Gas Equivalencies Calculator https://www.epa.gov/energy/greenhouse-gas-equivalencies-calculator, last accessed 2021/12/12.
18. OECD Environment at a Glance. https://www.oecd-ilibrary.org/docserver/9789264185715-en.pdf?expires=1666346904&id=id&accname=guest&checksum=05F8F534AFE7383B64E1E3CF7E12DCB2, last accessed 2021/12/12.
19. Eurostat. Electricity price statistics for non-household consumers. https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Electricity_price_statistics#Electricity_prices_for_non-household_consumers, last accessed 2022/7/9.

Open Access This chapter is licensed under the terms of the Creative Commons Attribution 4.0 International License (http://creativecommons.org/licenses/by/4.0/), which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license and indicate if changes were made.

The images or other third party material in this chapter are included in the chapter's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the chapter's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.